



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI VENKATESWARA FUELS UNIT II

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

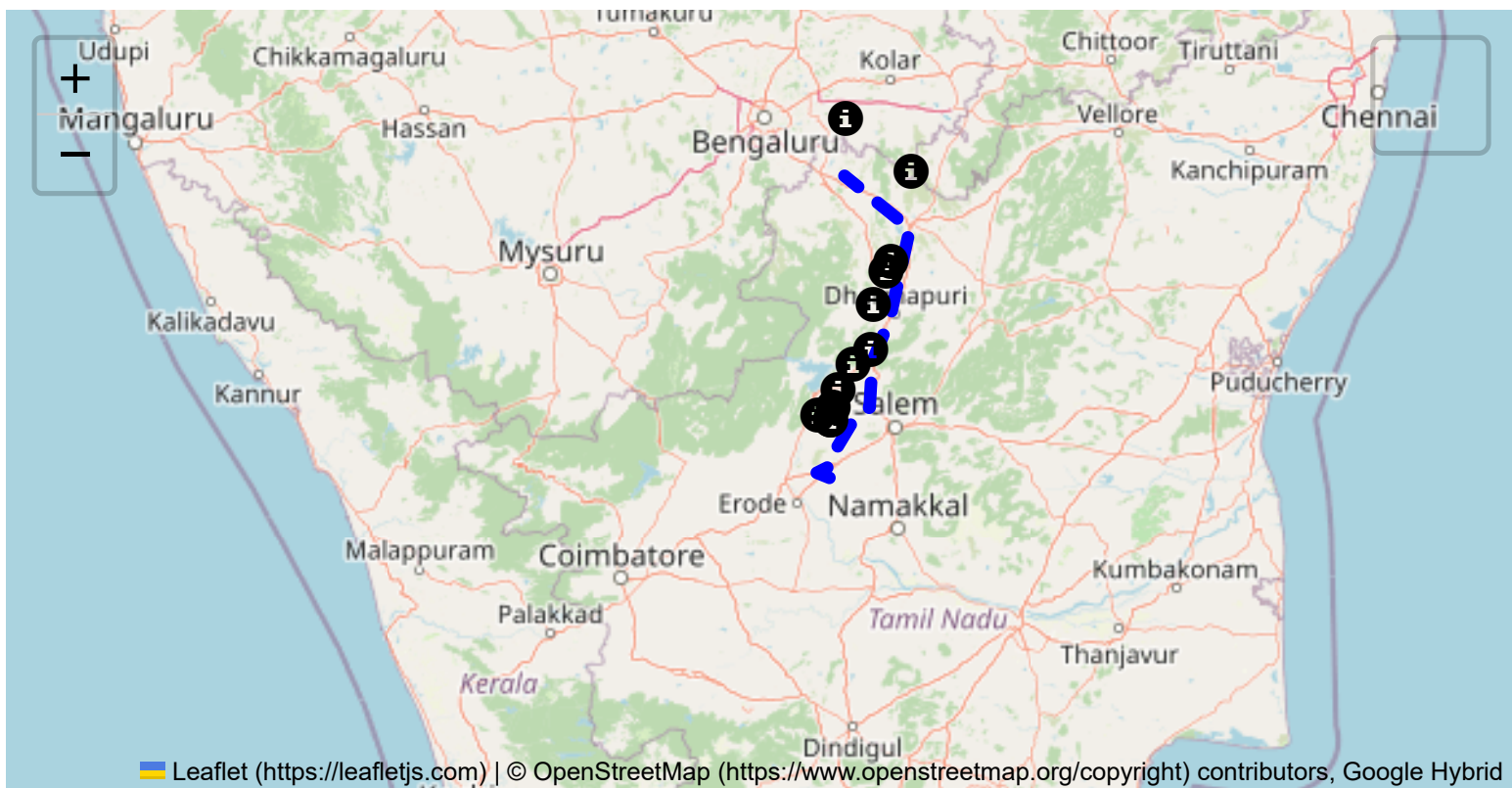
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 195.04 km
Estimated Duration: 3.8 hours
Adjusted Duration (Heavy Vehicle): 4.7 hours
Start: (11.4381, 77.8734)
End: (12.725437, 77.938927)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Krishnagiri, Tamil Nadu

1. Route Overview: The route spans approximately 195.04 kilometers through multiple key locations in Tamil Nadu, taking roughly 3.79 hours for heavy vehicles carrying hazardous materials. It passes through several towns and major roads, starting from Sangagiri and ending in Krishnagiri.

2. Weather Conditions:

- **Typical Weather:** Tamil Nadu experiences tropical weather with hot summers, moderate winters, and monsoon seasons.
- **Hazardous Weather:** Monsoon rains can lead to slippery roads, flooding in low-lying areas, and reduced visibility. Dry months may cause dust in certain areas, affecting visibility as well.

3. Traffic Patterns:

- **Peak Hours:** Generally, morning (7-9 AM) and evening (5-7 PM) are peak traffic hours. Expect congestion around major towns like Salem.
- **Congestion-Prone Areas:** Areas near Salem and Dharmapuri can experience heavy traffic, especially on highways and major junctions.

4. Road Quality and Infrastructure:

- **General Assessment:** Roads are generally well-maintained in populous areas but may become narrow and less maintained in rural segments.
- **Concerns:** Watch for potholes, unexpected debris, and roadworks, particularly on less traveled roads between smaller towns.

5. Alternative Routes for Emergencies:

- **Via NH 44:** Allows bypassing some smaller towns by sticking to major highways, if there is an issue in one of the towns along the original route.
- **Via SH 86 and NH 179A:** Offers parallel paths, especially useful when major routes are closed due to accidents or weather.

6. Local Regulations:

- Heavy and hazardous materials require compliance with Tamil Nadu's transportation regulations, including specific lane use and speed restrictions, especially in urban limits.

7. Historical Incidents:

- **Heavy Vehicles:** Past incidents often occur during monsoon seasons due to slippery roads or mechanical failure. Over-speeding and overloading are common contributing factors.
- **Hazardous Materials:** Any incidents often involve improper handling or failure to adhere to safety protocols.

8. Environmental Considerations:

- **Sensitive Areas:** Proximity to agricultural lands means potential spill impact could be significant. Wildlife regions near Dharmapuri require careful transit to avoid impacting biodiversity.

9. Communication Coverage:

- **Potential Dead Zones:** Rural segments, especially around forested areas between towns, may experience poor mobile network coverage. It's advisable to have a satellite communication backup.

10. Emergency Response Times:

- **Urban Areas:** Quick response due to proximity to towns' emergency services.

- **Rural Segments:** Expect longer response times, potentially up to an hour, especially in remote stretches.

11. Summary of Risk Assessment:

- **Overall Risks:** Moderate, with potential issues arising from weather, road quality, and traffic congestion.
- **Recommendations:** Stay updated on weather forecasts, follow local traffic advisories, and maintain communication with local authorities. Regular vehicle checks, especially brakes and tires, are crucial for safe transport of hazardous materials.

Navigating this route requires vigilance, adherence to traffic rules, and constant updates from local transport authorities to mitigate risks effectively. Safe driving practices and preparation for emergency situations are paramount for ensuring the safety of both the driver and the environment.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44077, 77.87533	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	Medium	11.45352, 77.85698	30 KM/Hr
4	Turn	High	11.46303, 77.84945	15 KM/Hr
5	Turn	Medium	11.45545, 77.81812	30 KM/Hr
6	Turn	High	11.45584, 77.81695	15 KM/Hr
7	Turn	Medium	11.45656, 77.81709	30 KM/Hr
8	Turn	High	11.49448, 77.88490	15 KM/Hr
9	Turn	Medium	11.56352, 77.90078	30 KM/Hr
10	Turn	Medium	11.59034, 77.90578	30 KM/Hr
11	Turn	Medium	11.59328, 77.90809	30 KM/Hr
12	Turn	Medium	11.63867, 77.94841	30 KM/Hr
13	Turn	High	11.64217, 77.95200	15 KM/Hr
14	Turn	Medium	11.64712, 77.95118	30 KM/Hr
15	Turn	Medium	11.67858, 77.96834	30 KM/Hr
16	Turn	Medium	11.69776, 77.97983	30 KM/Hr
17	Turn	Medium	11.70179, 77.97900	30 KM/Hr
18	Turn	Medium	11.71394, 77.99572	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
19	Turn	Medium	11.71916, 78.01117	30 KM/Hr
20	Turn	Medium	11.72504, 78.01680	30 KM/Hr
21	Turn	Medium	11.73766, 78.03445	30 KM/Hr
22	Turn	High	11.73651, 78.03868	15 KM/Hr
23	Turn	Medium	11.74120, 78.04588	30 KM/Hr
24	Turn	Medium	11.86805, 78.08353	30 KM/Hr
25	Turn	Medium	11.88084, 78.08108	30 KM/Hr
26	Turn	Medium	11.91159, 78.05857	30 KM/Hr
27	Turn	Medium	11.93307, 78.06426	30 KM/Hr
28	Turn	Medium	11.93545, 78.06130	30 KM/Hr
29	Turn	Medium	11.93694, 78.05425	30 KM/Hr
30	Turn	Medium	11.95648, 78.04768	30 KM/Hr
31	Turn	High	11.95803, 78.04964	15 KM/Hr
32	Turn	Medium	11.96068, 78.04960	30 KM/Hr
33	Turn	High	11.96370, 78.05142	15 KM/Hr
34	Turn	High	11.96650, 78.04927	15 KM/Hr
35	Turn	Medium	12.12094, 78.13269	30 KM/Hr
36	Turn	Medium	12.14025, 78.12819	30 KM/Hr
37	Turn	High	12.14099, 78.12862	15 KM/Hr
38	Turn	High	12.14056, 78.13957	15 KM/Hr
39	Turn	Medium	12.17102, 78.16860	30 KM/Hr
40	Turn	High	12.24611, 78.19519	15 KM/Hr
41	Turn	Medium	12.25200, 78.20034	30 KM/Hr
42	Turn	Medium	12.31895, 78.20727	30 KM/Hr
43	Turn	Medium	12.40026, 78.21775	30 KM/Hr
44	Turn	Medium	12.40734, 78.22344	30 KM/Hr
45	Turn	Medium	12.50342, 78.22352	30 KM/Hr
46	Turn	Medium	12.54446, 78.20114	30 KM/Hr
47	Turn	Medium	12.61917, 78.08303	30 KM/Hr
48	Turn	Medium	12.63696, 78.07061	30 KM/Hr
49	Turn	Medium	12.67143, 77.97400	30 KM/Hr
50	Turn	Medium	12.67742, 77.96591	30 KM/Hr
51	Blind Spot	Blind Spot	12.68367, 77.93734	10 KM/Hr
52	Turn	High	12.68387, 77.93773	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
53	Turn	Medium	12.68704, 77.93806	30 KM/Hr
54	Turn	Medium	12.69087, 77.93531	30 KM/Hr
55	Turn	Medium	12.70172, 77.93481	30 KM/Hr
56	Turn	Medium	12.70536, 77.93250	30 KM/Hr
57	Turn	Medium	12.71441, 77.93401	30 KM/Hr
58	Turn	Medium	12.71572, 77.93559	30 KM/Hr
59	Turn	High	12.72519, 77.93605	15 KM/Hr
60	Turn	Medium	12.72574, 77.93731	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
13	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
14	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
15	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
22	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
23	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
25	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
29	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
30	clinic	Joe Hospital	12.5209601, 78.2150623	30 km/h	Medium
31	clinic	Ohm Prakash Clinic	12.5206321, 78.2159682	30 km/h	Medium
32	clinic	Gokulam Clinic & Scans	12.5207976, 78.2162023	30 km/h	Medium
33	clinic	Srinivasa Clinic	12.520975, 78.215755	30 km/h	Medium
34	clinic	Dr. Anbalagan Clinic	12.5208366, 78.2160593	30 km/h	Medium
35	clinic	Shree Ramachandra ENT Centre	12.5290096, 78.2037359	30 km/h	Medium
36	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
37	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
38	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium
43	pharmacy	Mithran Medicals	12.6837657, 77.93838	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
24	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
26	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
27	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
28	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
39	college	Government Medical College	12.582332, 78.1496363	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
40	school	Crescent Matriculation Higher Seconddary School	12.6003724, 78.132787	30 km/h	Medium
41	college	Jeeva College of Nursing	12.6067058, 78.1191944	30 km/h	Medium
42	school	New Little Flower Matriculation Higher Secondary School	12.6607923, 78.0169012	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44077, 77.87533



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45545, 77.81812



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45584, 77.81695



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 77.81709



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49448, 77.88490



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56352, 77.90078



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59034, 77.90578



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59328, 77.90809



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69776, 77.97983



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70179, 77.97900



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71916, 78.01117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72504, 78.01680



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73766, 78.03445



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74120, 78.04588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86805, 78.08353



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.88084, 78.08108



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91159, 78.05857



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93307, 78.06426



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93545, 78.06130



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93694, 78.05425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95648, 78.04768



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95803, 78.04964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96068, 78.04960



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96370, 78.05142



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14025, 78.12819



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14099, 78.12862



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14056, 78.13957



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17102, 78.16860



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.24611, 78.19519



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25200, 78.20034



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31895, 78.20727



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40026, 78.21775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40734, 78.22344



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50342, 78.22352



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.54446, 78.20114



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.61917, 78.08303



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.63696, 78.07061

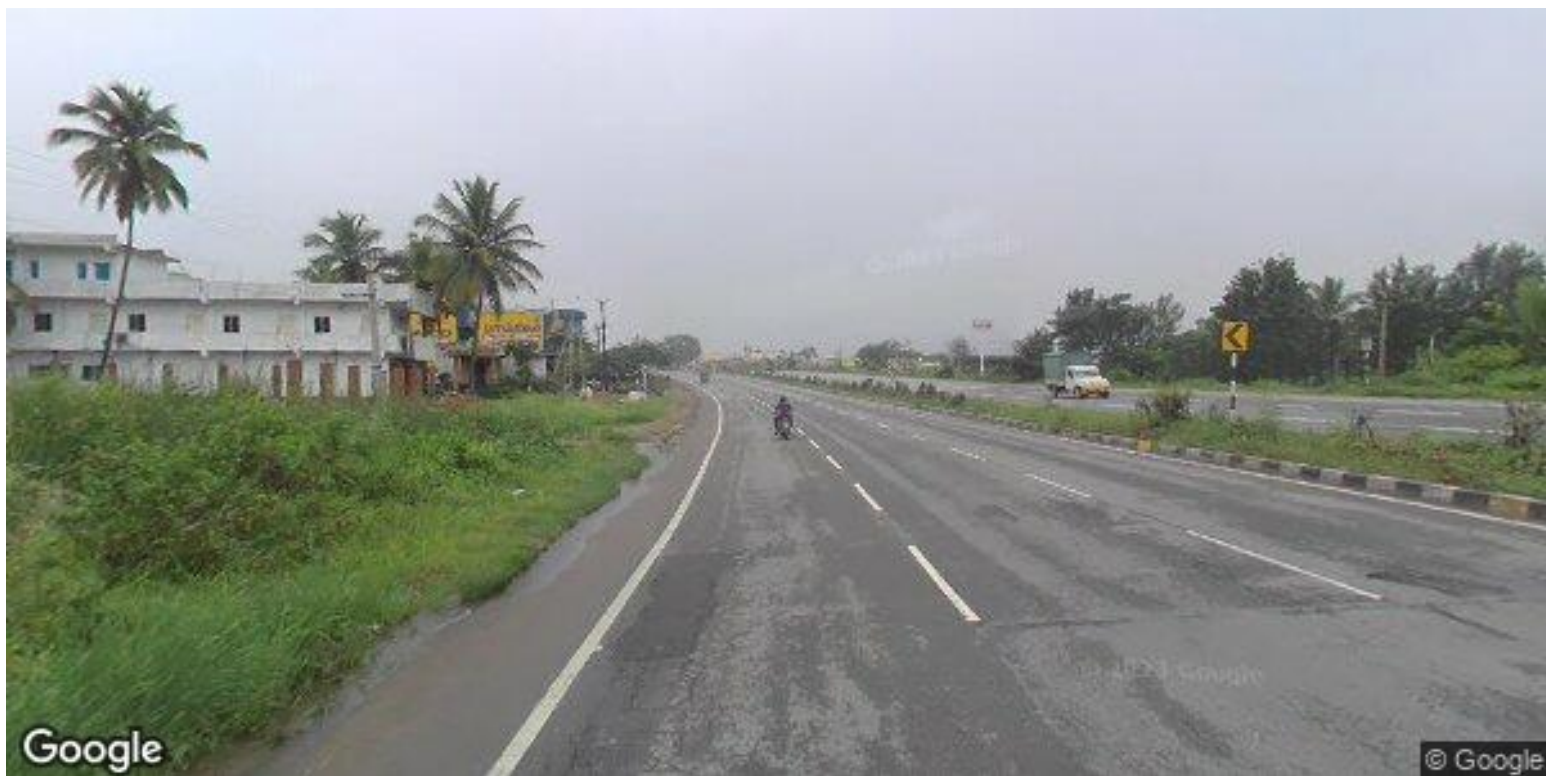


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.67143, 77.97400



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.67742, 77.96591



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.68367, 77.93734



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.68387, 77.93773



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.68704, 77.93806



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.69087, 77.93531



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.70172, 77.93481



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.70536, 77.93250



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.71441, 77.93401



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.71572, 77.93559



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.72519, 77.93605



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.72574, 77.93731
